

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Originating Company	Epic Atlantic Limited
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Specification for Piping Valves

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Rev No.	Date	Description of Revision
R01	24-05-22	First Issue for Clients Review
A02	16-12-22	Issued for Approval

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

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Abbreviations

Abbreviation	Meaning
AED	Anti-Explosive Decompression
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CS	Carbon Steel
ENP	Electroless Nickel Plated
ISO	International Standard Organisation
MESC	Material & Equipment Standard and Code
MMscf/d	Million standard cubic feet per day
MSS	Manufacturers Standardisation Society
MT0	Material Take Off
NDE	Non-Destructive Examination
NPT	Nominal Pipe Threaded
NPS	Nominal Pipe Size
SPE	Society of Petroleum Engineers
SS	Stainless Steel
SW	Socket Weld

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

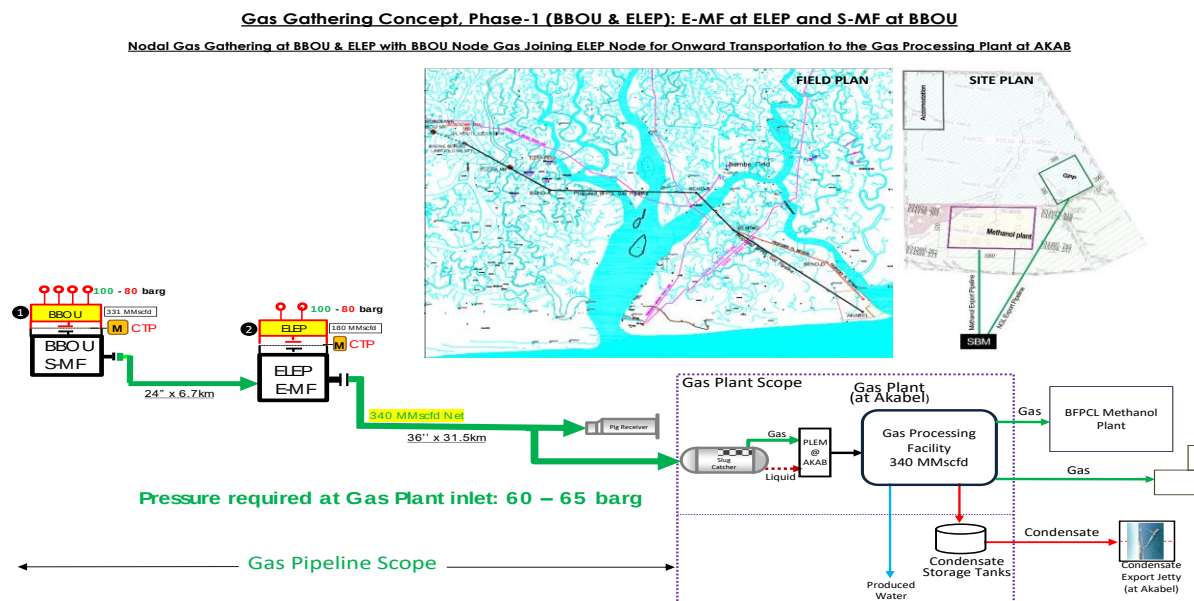


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum requirements for design, material selection, testing, inspection, surface preparation and packaging of manual and gear operated valves that shall be installed within area design to ASME B31.3 or piping scope in for BFPCL Phase 1 Project.

The valves to be designed, manufactured, tested and supplied by the Vendor are those listed in piping Valve MTO referenced in the valve Data sheets and Requisitions.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.11	Piping Classes – Refining and Chemicals
DEP 30.10.01.11	Metallic Materials – Selection and Application
DEP 30.10.01.13	Non-metallic Materials – Selection and Application
MESC SPE 77/302	Valves – General requirements
SPE 77/103	Globe Valve To BS 1873
SPE 77/104	Check Valve To BS 1868
SPE 77/130	Ball Valve to ISO 14313 / API 6D, Flanged or Butt Weld Ends
ASME B1.20.1	Pipe Threads, General Purpose (Inch)
ASME B16.5	Pipe Flanges and Flanged Fittings NPS ½ Through NPS 24
ASME B16.10	Face-to-Face and End-to-End Dimensions of Valves
ASME B16.34	Valves – Flanged, Threaded, and Welding End
ASME B31.3	Process Piping
ASTM A703	Standard Specification for Steel Castings, General Requirements, for Pressure-Containing Parts
ASTM B 633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
API Spec 6D	Specification for Pipeline Valves
API Spec 6 FA	Specification for Fire Test for Valves
API Spec 6B	Specification for Fire Test for End Connections
API 594	Check Valves: Flanged, Lug, Wafer and Butt-Welding
API 598	Valve Inspection and Testing
API 602	Steel Gate, Globe and Check Valves for Sizes NPS 4 (DN100) and Smaller for the Petroleum and Natural Gas Industries
API 607	Fire Test for Quarter-Turn Valves and Valves Equipped with Non-metallic Seats
ISO 14313	Petroleum and natural gas industries – Pipeline transportation

	systems – Pipeline valves
MSS SP 25	Standard Marking System for Valves, Fittings, and Unions
MSS SP 55	Quality Standard for Steel Castings for Valves, Flanges, Fittings, and Other Piping Components- Visual Method for Evaluation of Surface Irregularities.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Design Data

Refer to Project Basis of design document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002.

4. Requirements

4.1. General Requirements for Valves

Valves shall be designed and engineered for a service life of 30 years.

All valves shall be designed in accordance with API 6D and MESC SPE 77/302. The design and calculations for pressure-containing elements shall be in accordance with ASME B16.34.

Additional project specific requirements shall be listed on the valve data sheets and the material requisitions.

All valves shall be Fire-safe tested and certified in accordance with ISO 10497, S 6755 part 2, API 607 or API 67A. Fire-safe design shall be graphitic or metallic sealing arrangement capable of passing fire test to API 607 and API 6FB.

Where Tight shut-off is specified, valve shall comply with seat leakage rates A or B according to ISO 5208 and S 6755-1.

All valves shall be suitable for vacuum drying and vacuum service at 5 mbar abs.

The valve body shall be designed with sufficient rigidity to withstand a compressive axial thrust exerted by the mating pipe, equal to the shell test pressure multiplied by the cross sectional area of the mating pipe, without permanent distortion of internals affecting the free movement and sealing of the obturator and the seats.

Lifting lugs shall be provided for valves with mass of 250kg and above or for valves with size more than 10 inches. Valve support material shall be of same type as for the valve body

Valve supports (legs or saddles) shall be supplied and installed on valves of nominal size 12 inches and above. The supports must allow access to body drain plugs where fitted.

Flanged ends shall be furnished with Standard raised face (RF) and Ring Type Joint (RTJ) in accordance with ASME B16.5. The flange facings shall undergo finish machining in accordance with MSS SP-6

If welding ends are specified; it shall be furnished with extension pieces which shall be considered as an integral part of the valve. Length of pup piece shall be 100mm for sizes less than 2 inches, 400mm for sizes $\leq$ 12 inches and 800mm for sizes more than 12 inches. The outside diameter, wall thickness, material grade and composition shall conform to the adjoining pipe specification.

All manual valves (whether operated with a lever or a gear operator) shall be provided with locking device plate to facilitate padlocks or car-seals.

All valves shall have position indicator and the design shall be such that the component(s) of the indicator and the valve operator cannot be assembled to falsely indicate the valve position.

Sealant injection facilities for upper stem and seat seals shall incorporate a check valve and a secondary means of isolation. All connections shall be butt-welded to the valve body, or alternatively, shall be seal welded if it is a screwed connection

Valve stem design shall be of blowout-proof type. Note that ball valve stem retention shall not depend on the packing gland: a bottom entry shouldered stem is required.

The maximum dimensions of hand-wheel and levers shall be in accordance to the requirements of API Spec 6D / ISO 14313 except that the maximum hand wheel diameter shall be limited to 800 mm and the maximum lever length shall be limited to 450 mm.

The maximum force required under design and test conditions, when applied at the rim of the hand-wheel or lever, shall not exceed 300 N during valve opening or closing, and 400 N when unseating the valve. This applies to seating, unseating, and operating at maximum pressure differential for both minimum operating temperature and ambient temperatures. If the force or dimensional limitations are exceeded, the valve shall be provided with a gearbox.

Gear operators shall be dust-proof and weather-proof by total enclosure. Gearboxes shall be sealed, and lubricated for life, and suitable for operating at an ambient temperature of +38°C, fitted with a gearbox mounting flange in accordance with ISO 5211.

Valves used in fluids containing solid particles shall have metal-to-metal seating.

Marking on valves shall be as per MSS SP 25

Except for stainless steel materials, bolting shall be provided with zinc plated / bichromate surface treatment (ISO 4042 - 12 C Fe or ASTM F 871 and ASTM B 633 - Fe Zn 12 Sc3).

In addition to name plate, valve shall be supplied with a stamped SS316 tag fixed to the gland bolting or handwheel with SS316 wire indicating project information such as PO No., tag number, valve type, size, rating, painting system, item number reference to valve MTO attached to RFQ, etc.

Face to Face and End to End dimensions and full and reduced bore sizes of all valves shall comply with API Spec 6D / ISO 14313. However, ASME B16.10 is applicable for all valves or sizes not listed in the API Spec 6D / ISO 14313.

The end-to-end dimension of threaded or socket-welding valves shall be indicated by equivalence to a Manufacturer's catalogue.

Valves not covered by a standard (e.g. Integrated DBB Valves, Monoblock Valves) shall be per manufacturer's standard and shall be considered a specialty item.

4.2. Requirements and Limitation for Materials

Valve material selection shall be as specified in the valve datasheets and the projects piping material specification and classification.

No specified material substitution with an equivalent or higher material shall be allowed without a written authorization from EMPLOYER.

Cast material cannot be proposed as substitute to forged material unless specifically agreed by the EMPLOYER. Unless otherwise specified steel casting for pressure containing parts shall comply with general requirements of ASTM A703 with all its supplementary requirements.

Cast material shall not be used for valve stem whatever the pressure class and valve size.

It is the responsibility of the Manufacturer to ensure that elastomeric seals selected are Anti Explosive Decompression certified and is compatible with the process and operating conditions of the valve

Graphite seals are not acceptable as primary pressure containment seal. They shall only be used as back up seal for fire resistant properties.

The chemical composition and carbon content restriction for valve pressure containing parts (body, bonnet, cover) and stem shall be as detailed in MESC SPE 77/302.

Carbon steel forgings to ASTM A105 shall be furnish in the normalised condition.

All static and dynamic sealing surfaces of carbon steel intended for use with lip seals shall be weld-over layed with grade 316 stainless steel or Alloy 625 material per SPE 77/313. Surface finish shall be $Ra < 0.2\mu m$.

Hard facing where specified shall be an Electroless Nickel Plating treatment (ENP) of 0.003 inch (75 μ) minimum thickness in accordance with a certified manufacturer's procedure. The coated surfaces shall be 100% visually examined and shall be free from pores and cracks.

4.3. Technical Requirements for Ball Valves

4.3.1. General

- Ball valve design shall comply with the requirements of MESC SPE 77/302 as a supplement to API 6D and MESC SPE 77/130
- Ball valves shall normally be reduced bore, unless otherwise specified on the data sheet and requisition.
- Valve installed on piggable line shall be full bore valve with same bore all through the valve. The bore shall match the pipeline internal diameter. Full bore valves shall be capable of being pigged regularly without damage to the soft seat material (if fitted).
- The obturator shall be a one piece forging or casting (if approved by purchaser) and the valve port shall be cylindrical.
- Ball valve shall be bi-directional and shall be fully dismantled bolted design, top or side entry as indicated in datasheet. For welded end valve, only top entry design is allowed.
- Short pattern ball valves shall not be furnished.
- Ball valves shall be trunnion mounted, floating type, if manually operated, are acceptable in the following case:
 - ✓ CL 150: Diameters $\leq 6"$
 - ✓ CL 300: Diameters $\leq 4"$

✓ CL 600, 900, 1500: Diameters $\leq 1\frac{1}{2}$ "

- Actuated ball valves shall be trunnion mounted exclusively, whichever their size or pressure Class.

4.4. Design Requirements

4.4.1. Anti-Static Design

- Ball valves are required to be of Anti-Static Design. Anti-static design shall incorporate suitable devices to ensure electrical continuity between the ball, stem and the body. Graphite seals shall not be considered to offer satisfactory electrical continuity.

4.4.2. Drain Connections

- Connection for drain where specified in the datasheet shall be drilled and threaded. The thread shall have a parallel profile in accordance with ISO 228-1 and fitted with a parallel threaded plug.
- Thread/pipe sizes for drains shall be in accordance with Table below:

Nominal Pipe Size	Minimum pipe thread/pipe size
2 – 8 inches	15mm (1/2")
Above 8 inches	25mm (1")

- Drains, vents and sealant points, which may have been included for testing, shall be blanked off.

4.4.3. Cavity Pressure Relief

- Pressure relief shall be by means of self-relieving seat rings or a pressure relief valve connected to the cavity for valves with double piston effect seating. Pressure relief shall not be achieved by using a pressure-equalising hole in the obturator connecting the body cavity to the high pressure side of the closure member. Manufacturer shall provide a manual that explains how pressure relief operates and how to assemble closure members.

4.4.4. Design of seat rings

- Seat rings shall be renewable and shall be mechanically energized to obtain positive sealing under low pressure differential. Under high differential pressure, it shall be line-pressure energized.
- The design of the seat ring and valve body shall be such that entrapment of any solids present in the fluid shall not obstruct the free movement of the seat rings and shall not reduce spring action.

4.5. Technical Requirements for Globe and Check Valves

The design and construction of globe and check valves shall comply with the requirements of BS 1873 and supplemented by MESC SPE 77/103

Globe valve shall be of Outside Screw and Yoke (OS&Y), rising stem rising operator (RSRO), and bolted bonnet design, straight pattern.

Globe valves shall have threaded and removable seats and shall have swivel plug disc.

Globe valve stem design shall allow for tightness at intermediate obturator position when the valve is intended for flow control.

Only dual plate check valves of retainerless design shall be used. Dual check valves if specified shall be designed as per API 594.

The body to bonnet connection shall be flanged and the flange facings shall be male-and-female, tongue-and-groove. Flange sealing faces shall have a circular shape and shall have a facing finish in accordance with ASME B16.5

For carbon steel swing type check valves, disc retaining components e.g. studs, nuts, washers and pins shall be of the same material as specified for the "trim" of the valve.

Wafer-lug type dual plate check valves will be of the type where the bolting is continuously covered across the valve body by the lug, thereby protecting the bolting from direct fire impingement (in the event of a fire) or degradation by general corrosion.

5. Valve Inspection and Testing

5.1. General

The fire test certificate shall be submitted with the general arrangement drawing and the complete part list of the valve that was qualified as to validate the "fire safe" design of the supply.

During the valve Factory Acceptance Test, the EMPLOYER or its representative shall have free access to the valve in the bunker or on the test bench before and after each individual step of the valve test sequence, regardless any remote monitoring system used to follow the test performance.

The FAT, Factory Acceptance Test, of valves shall be notified by the Manufacturer only after successful internal (in-house) testing. The internal test reports shall be made available to the Inspection Agency prior to any further consideration for the related Notice For Inspection (NFI).

Operation of the actuator shall be smooth at all times along the full stroke. Any kind of unexpected actuator movement during the valve stroke test shall be identified and reported.

5.2. Detailed Inspection

5.2.1. Visual examination

All valves must be 100% visually inspected as per API 598 and MSS SP 55.

5.2.2. Special examination

All non-destructive examination on casting is deemed performed on finished casting after all repair works and the complete heat treatment sequence.

For all valves 6" and larger in pressure class 600 and higher:

- Castings shall be examined in accordance with section 8 of ASME B16.34 (Requirements for special class valves, examination of steel castings) and MSS SP 55.
- The extent of radiographic examination of cast components shall also include location of abrupt change in geometry or thickness, location of risers, gates and feeders.
- The following shall apply to components of lower pressure ratings:
 - ✓ 10% of all valves of size 6" to 18" class 150# and 300#.
 - ✓ 20% of all valves 20" and larger 150# and 300#.
- A full radiographic inspection shall be carried out on 100% of the bevelled areas of all weld-end valves of sizes 2" and above, and on all butt-welded joints, to prove that these areas are free from any defects. Bevelled area of valve ends made of forgings may be inspected by ultrasonic testing instead of radiography.
- Any weld (including fillet welds) on valve bodies, regardless of size or pressure class, shall be 100% examined by dye penetrant testing.

5.3. Detailed Test

All valves shall be 100% tested by the Manufacturer in his shop and under his responsibility.

The Manufacturer shall submit for the EMPLOYER's approval its detailed testing procedure for each type of valves, including each pressure class and size. This procedure must be in full accordance with the present specification requirements and the applicable codes.

5.3.1. Pressure Test

- All valves (i.e., 100% of the supply) shall undergo pressure tests in accordance with the requirements of API Spec 6D and API 598. Valves not covered by these standards shall be tested in accordance with MSS SP 61.
- For valve types other than ball valve the test shall be identified by specific valve marking. Optional tests that are described in table 1 or table 2 of API 598 become mandatory irrespective of valve size and valve pressure class.
- Back seat test of globe valves used in gas service shall be performed with nitrogen at 110% the pressure rating of the valve.

5.3.2. Test medium

- Testing medium shall be clean, inhibited, fresh water. Stainless steel valves shall be tested using inhibited fresh water having a maximum chloride content of 15 ppm. This also applies to valves fabricated with some parts made of stainless steel. The temperature of the test must be a minimum of +10°C. In case of non-impact tested body materials, the minimum temperature of the test medium shall be +16°C.
- After testing with water, all valves shall be thoroughly dried to prevent possible corrosion from the water.
- High-pressure hydrostatic seat test leakage rates shall comply with API Spec 6D

5.3.3. Gas tests

- Gas tests shall be monitored using soap bubble method where zero leaks are either expected or requested. Where leak rates other than zero are specified, the leakage rate shall be measured using a flow meter calibrated with the gas used for testing.

5.3.4. Toughness tests

- The toughness tests to be carried-out on the valves (if required) are normally shown on the Purchase Order documents.
- When these tests are required, they shall be carried out in conformity with the requirements of ASTM A 370 and at the temperature required for the grade of material.

5.3.5. Hardness tests

- Hardness tests shall be carried out on each valve body (e.g. for compliance with the NACE standards in case of H2S service).

5.3.6. Test failure, re-work, re-test

- In case a test fails, the Manufacturer is permitted to re-submit the same valve only once, and only after it has been properly re-worked and the reasons for failure thoroughly examined and explained by the Manufacturer to the Inspection Agency (who must report them). Should a given valve fail to pass a test for the second time, it shall be rejected and replaced by a new one.

5.4. Painting

Surface preparation and application of painting and protective coatings shall be carried out in accordance with Painting and Coating for Above Ground Specification for the project. Handle or steering shall be in the same material grade as the valve body to avoid galvanic corrosion.

The interior surfaces, the threaded parts, the bevelled or socket-welding ends shall be painted if not machined. The contact seal surfaces and valve tag identification plate shall not be painted.

5.5. Preservation and Shipping

Machined surfaces shall be cleaned with a suitable solvent and immediately protected with an easily removable rust preventative coating or plastic film.

One-piece marine grade plywood or heavy plastic caps shall be taped to flanged connections.

The Vendor shall provide a preservation specification including recommendations to be used at yard and at site for long duration storage.

Vendor shall include details of recommended protective and rust preventive coatings required for preservation of their valves. Valves shall be properly preserved, fitted with end caps or flange protectors, placed on an adequate surface and packaged to ensure protection against damage, ingress of moisture, rainfall and dirt during outdoor storage for at least one year.

5.6. Documentation

Manufacturer shall provide documentation prior to shipment and shall comprise of the following as a minimum;

- It shall be in the English language only.
- It shall be in SI units, although flange diameters shall be in nominal inch size.
- Drawings and other documentation shall be in A4 or A3 size as appropriate.
- Test reports showing locations of examination traceable by heat or serial number.
- Material test reports and inspection certificates, traceable by heat number to the foundry or mill.

The full vendor documentation requirement shall be detailed in the requisition

5.7. Certification

In addition to the certification requirements specified in MESC SPE 77/302, the finished valve

shall have a certificate in accordance with ISO 10474 type 2.1 or EN 10204 type 2.1, demonstrating that it complies with all requirements of API 6D or other referenced design standards. The certificate(s) shall identify the valve type, size, class, end connection, serial number(s), and any additional requirement specified by the manufacturer.